Activity 1: Device Physics Evidence Map

Course: Fundamentals of Semiconductor Device Physics and Process Integration

Course code: TGS-2024049339

Duration: 35 minutes

TSC alignment: K1 | A1

PPT alignment: Trainer Slides v7.0, concept slides 18-105 and Activity 1 overview slide 441.

Workplace scenario

A new process engineer must explain why doping, junction bias and oxide thickness matter before reviewing a CMOS traveller.

Goal

Connect device-physics mechanisms to observable manufacturing controls and device consequences.

Files in this folder

- README.md / README.pdf - complete procedure and acceptance criteria.
- SCENARIO.md / SCENARIO.pdf - facts, assumptions and role brief.
- WORKSHEET.md / WORKSHEET.pdf - structured learner response.
- EVIDENCE-CHECKLIST.md / EVIDENCE-CHECKLIST.pdf - completion and review record.
- Any CSV file in this folder - supplied teaching data for analysis.

Step-by-step procedure

1. Review the supplied device cross-sections and identify the semiconductor, oxide, junction and contact regions.
2. For each region, state the governing mechanism: carrier concentration, depletion, inversion, recombination or field-driven transport.
3. Map each mechanism to one controllable process input such as dose, energy, anneal temperature, oxide thickness or critical dimension.
4. Predict one electrical symptom when that input drifts high and one when it drifts low.
5. Mark the evidence that would confirm the prediction and record one uncertainty requiring follow-up.

Required evidence

Completed device-physics evidence map with at least six mechanism-control-consequence links.

Include your completed worksheet, any calculations or annotated diagrams, the evidence checklist, and a short note identifying assumptions or unresolved evidence gaps.

Acceptance criteria

[ ] All four device regions are addressed

[ ] At least six causal links are technically plausible

[ ] Each prediction names a measurable symptom

Verification

Before submission, ask a peer to challenge one causal link. Revise the conclusion if the challenge reveals that evidence and inference have been mixed. Your final response must distinguish: observed fact, interpretation, decision, and follow-up check.

Troubleshooting

- If several mechanisms fit the symptom, choose a measurement that discriminates between them.
- If a process module lacks a defined input, output or control gate, the flow is incomplete.
- If an action is not tied to a verified cause, record it as a hypothesis test rather than corrective action.
- If calculations disagree, show the method and units so the discrepancy can be reviewed.